

# Tarea Esperanza de la Distribución Uniforme.

$$M_x(t) = E[e^{tx}]$$

$$= \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_a^b e^{tx} f(x) dx$$

$$= \int_a^b e^{tx} \left(\frac{1}{b-a}\right) dx = \frac{1}{b-a} \int_a^b e^{tx} dx$$

$$= \frac{1}{b-a} \left(\frac{1}{t}\right) \left(e^{tx}\right) \Big|_a^b = \frac{1}{t} \left(\frac{1}{b-a}\right) (e^{tb} - e^{ta})$$

$$\Rightarrow E[x] = \frac{d}{dt} \left[ \left(\frac{1}{b-a}\right) \left(\frac{1}{t}\right) (e^{tb} - e^{ta}) \right] \Big|_{t=0}$$

Como  $e^{tx} = \sum_{n=0}^{\infty} \frac{(tx)^n}{n!} \Rightarrow$

$$E[x] = \frac{d}{dt} \left[ \left(\frac{1}{b-a}\right) \left(\frac{1}{t}\right) \left( \sum_{n=0}^{\infty} \frac{b^n t^n}{n!} - \sum_{n=0}^{\infty} \frac{a^n t^n}{n!} \right) \right] \Big|_{t=0}$$

$$= \frac{d}{dt} \left[ \left(\frac{1}{b-a}\right) \left(\frac{1}{t}\right) \left( \sum_{n=0}^{\infty} \frac{(b^n - a^n) t^n}{n!} \right) \right] \Big|_{t=0}$$

si  $n=0 \Rightarrow b^0 - a^0 = 0$

$$= \frac{d}{dt} \left[ \left(\frac{1}{b-a}\right) \left(\frac{1}{t}\right) \left( \sum_{n=1}^{\infty} \frac{(b^n - a^n) t^n}{n!} \right) \right] \Big|_{t=0}$$

$$= \frac{d}{dt} \left[ \left(\frac{1}{b-a}\right) \left( \sum_{n=1}^{\infty} \frac{(b^n - a^n) t^{n-1}}{n!} \right) \right] \Big|_{t=0}$$

$$= \frac{d}{dt} \left[ \frac{1}{b-a} \left( (b-a) + \frac{(b^2 - a^2)}{2} t + \sum_{n=3}^{\infty} \frac{(b^n - a^n) t^{n-1}}{n!} \right) \right] \Big|_{t=0}$$

$$= \frac{1}{b-a} \left( 0 + \frac{(b^2 - a^2)}{2} + \sum_{n=3}^{\infty} \frac{(n-1)(b^n - a^n)}{n!} t^{n-2} \right) \Big|_{t=0}$$

$$= \frac{1}{b-a} \left( \frac{b^2 - a^2}{2} \right) = \frac{1}{b-a} \left( \frac{(b+a)(b-a)}{2} \right)$$

$$= \frac{b+a}{2}$$